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Quiz on Public Key Cryptography
Graded Assignment • 20 min

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Introductory Material

Introduction to Public Key Cryptography

Video: Introduction to Public Key Cryptography

20 min

Graded Assignment: Quiz on Public Key Cryptography

Started

GCD, Euclid's Algorithm and Bezout Coefficients

RSA Public Key Cryptosystem

Basic Introduction to Quantum Computation

Problem Set: Week 1

Your grade: 100%

Next item →

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

100%

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Review Learning Objectives

Assignment details

2 / 2 points

1.

Suppose Alice wishes to send a message to Bob legally agreeing to purchase his oil well for \$25,000,000. She download's Bob's public key, encrypts her message and sends it over the channel. Suppose Charlie is snooping over the channel and can read all of Alice's messages to Bob. Select all the true facts from the list below.

☐ Charlie can decrypt the message using Alice's public key.

☐ Upon receiving the message, Bob knows that it must have come from Alice.

☐ Charlie can decrypt the message using Bob's public key.

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☐ Charlie can decrypt the message using his own private key.

☒ Charlie cannot read the message unless he has knowledge of Bob's private key.

Correct

True

Due

Attempts

1

Try again

2.

Alice wishes to send a message to Bob wherein she does not care if Charlie reads the message but she would like Bob to know that it came from her and no one else.

☒ This is called the "authentication problem" in public key cryptography.

Correct

☐ Alice is interested in ensuring secrecy.

☒ Alice needs to encrypt the message with her own private key.

Correct

Indeed: then Bob can decrypt it with Alice's public key and then know that it must have come from her or whoever has her private key.

☐ Alice encrypts the message with Bob's public key

☐ Alice encrypts the message with her own public key.

3.

Suppose Alice wishes to send an agreement to Bob but wishes it to be secret. Bob will honor the agreement only if he can authenticate that the message came from Alice. Which of the following steps are needed to achieve both secrecy and authentication.

☒ Alice prepares a digest of the message (a hash value) that has a time stamp with the time of the message.

Correct

This is needed so that Alice cannot claim that her digital signature from a different message was tacked on to this one without her knowledge.

☐ Alice encrypts the digest using her own public key.

☒ Alice encrypts the digest using her private key.

Correct

Indeed

☐ Alice encrypts the message and digest with Bob's private key.

☐ Alice encrypts the message and digest with Bob's public key.

☒ Alice encrypts the digest with her private key, attaches it to the message.

Correct

Correct

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